Integration and Kinematics

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Integration & Kinematics Practice Exercises**

The diagram shows part of the curve $y = 6x - x^2$, which intersects the x-axis at $(0, 0)$ and $(6, 0)$. Find the area of the region enclosed between the curve and the x-axis.

Final Answer: 36. To find the area enclosed between the curve and the x-axis, integrate y with respect to x from $x = 0$ to $x = 6$: Area $= \int_0^6 (6x - x^2)\,dx$ Integrate term by term: $\int (6x - x^2)\,dx = \left[\frac{6x^2}{2} - \frac{x^3}{3}\right]_0^6 = \left[3x^2 - \frac{x^3}{3}\right]_0^6$ Substitute the limits: $\left(3(6)^2 - \frac{6^3}{3}\right) - 0 = 3(36) - \frac{216}{3} = 108 - 72 = 36$ Common errors: - Estimating the region as a triangle with.

2 **[Foundational]** • **Target: Indefinite Integration Techniques**

The curve C has equation $y = x^3 - 3x + 2$. The tangent to C at the point $P(2, 4)$ intersects the curve again at point Q. Find the area of the finite region completely enclosed between the curve C and the tangent line.

Final Answer: 108. To find the area of the enclosed region, we synthesize differential calculus, polynomial algebra, and definite integration: 1. **Find the equation of the tangent line at $P(2, 4)$:** Differentiating the curve equation: $\frac{dy}{dx} = 3x^2 - 3$ At $x = 2$, the gradient of the tangent is: $m = 3(2)^2 - 3 = 12 - 3 = 9$ Using the point-slope formula with $P(2, 4)$: $y - 4 = 9(x - 2) \Rightarrow y = 9x - 14$ 2. **Find the intersection point Q:** Set the curve and tangent equations equal.

3 **[Foundational]** • **Target: Definite Integrals and Area Under Curves**

A curve has equation $y = 6x - x^2$. The straight line $y = mx$, where $0 < m < 6$, intersects the curve at the origin and at the point P. Given that the area of the region enclosed between the curve and the line is $\frac{32}{3}$, find the value of the constant m.

Final Answer: 2. Step 1: Find the points of intersection of the curve and the line. Set the equations equal to each other: $6x - x^2 = mx \Rightarrow x^2 - (6 - m)x = 0 \Rightarrow x(x - (6 - m)) = 0$ Since the line intersects the curve at the origin $(0, 0)$ and at point P, the x-coordinate of P is $x = 6 - m$. Step 2: Set up the definite integral for the enclosed area. For $0 \le x \le 6 - m$, the curve lies above the straight line. The enclosed area A is given by.

© ClipSAT — clipsat.org | All Rights Reserved

4 [Foundational] • Target: Area Between Two Intersecting Curves

The diagram shows part of the curve $y = 9 - x^2$ for $x \geq 0$ and the tangent to the curve at the point $P(2, 5)$. Find the exact area of the finite region bounded by the curve, the tangent at P, and the x-axis.

Final Answer: $\frac{11}{24}$. 1. **Find the equation of the tangent at $P(2, 5)$:** Given $y = 9 - x^2$, differentiating with respect to x gives: $\frac{dy}{dx} = -2x$ At $x = 2$, the gradient of the tangent is $m = -2(2) = -4$. The equation of the tangent line is: $y - 5 = -4(x - 2) \Rightarrow y = 13 - 4x$ 2. **Determine the x-intercepts:** - Tangent: Set $y = 0 \Rightarrow 13 - 4x = 0 \Rightarrow x = \frac{13}{4} = 3.25$. - Curve: Set $y = 0 \Rightarrow 9 - x^2 = 0 \Rightarrow x = 3$ (since $x \geq 0$). 3..

5 [Foundational] • Target: Rectilinear Motion and Kinematics

The diagram shows the curve $y = 4x - x^2$. Find the area of the region enclosed by the curve and the x-axis.

Final Answer: $\frac{32}{3}$. To find the area enclosed by the curve and the x-axis, first find the x-intercepts by setting $y = 0$: $4x - x^2 = 0 \Rightarrow x(4 - x) = 0 \Rightarrow x = 0$ or $x = 4$. The area is given by the definite integral of y with respect to x from $x = 0$ to $x = 4$: $\int_0^4 (4x - x^2)\,\mathrm{d}x = \left[2x^2 - \frac{x^3}{3}\right]_0^4$ Evaluating between the limits.

6 [Foundational] • Target: Integration & Kinematics Practice Exercises

Find the area enclosed between the curve $y = 4x - 4$, the x-axis, and the lines $x = 1$ and $x = 4$.

Final Answer: 18. Area $= \int_1^4 (4x - 4)dx = [2x^2 - 4x]_1^4 = 18$

7 [Foundational] • Target: Indefinite Integration Techniques

Find the area enclosed between the curve $y = 4x - 8$, the x-axis, and the lines $x = 2$ and $x = 4$.

Area $= \int_2^4 (4x - 8)dx = [2x^2 - 8x]_2^4 = 8$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Definite Integrals and Area Under Curves

Two circles, C_1 with centre O_1 and radius 6 cm, and C_2 with centre O_2 and radius $6\sqrt{3}$ cm, intersect at the points A and B. The distance between their centres is $O_1O_2 = 12$ cm. Find the exact area of the region common to both circles, in cm^2.

Final Answer: $30\pi - 36\sqrt{3}$. 1. **Determine the angles subtended at each centre:** Consider triangle O_1AO_2 with side lengths $O_1A = 6$, $O_2A = 6\sqrt{3}$, and $O_1O_2 = 12$. Notice that: $(O_1A)^2 + (O_2A)^2 = 6^2 + (6\sqrt{3})^2 = 36 + 108 = 144 = 12^2 = (O_1O_2)^2$ By the converse of Pythagoras' Theorem, $\angle O_1AO_2 = \frac{\pi}{2}$ radians. Using right-angled trigonometry in ΔO_1AO_2.

© ClipSAT — clipsat.org | All Rights Reserved

9 [Intermediate] • Target: Area Between Two Intersecting Curves

The region R in the first quadrant is bounded by the curve $y = \frac{6}{\sqrt{3x+4}}$, the coordinate axes, and the vertical line $x = k$, where $k > 0$. Given that the total area of R is 8, and the vertical line $x = c$ (where $0 < c < k$) divides R into two regions of equal area, find the exact value of c.

Final Answer: $\frac{5}{3}$. Step 1: Integrate the function using the reverse chain rule. $\int 6(3x+4)^{-\frac{1}{2}}\,dx = \frac{6}{\frac{1}{2}\cdot 3}(3x+4)^{\frac{1}{2}} = 4\sqrt{3x+4}$ Step 2: Find the parameter k using the given total area of 8. Area $= \int_0^k \frac{6}{\sqrt{3x+4}}\,dx = \left[4\sqrt{3x+4}\right]_0^k = 4\sqrt{3k+4} - 4\sqrt{4} = 4\sqrt{3k+4} - 8$ Setting this equal to 8.

10 [Intermediate] • Target: Rectilinear Motion and Kinematics

Evaluate the definite integral: $\int_0^1 (9x+1)^3\,dx$.

Final Answer: $\frac{9999}{36}$. Substitute $u = 9x+1 \Rightarrow du = 9\,dx$. Integral: $\frac{1}{9}\left[\frac{u^4}{4}\right]_1^{10} = \frac{(10)^4 - 1^4}{36} = \frac{9999}{36}$.

11 [Intermediate] • Target: Integration & Kinematics Practice Exercises

Evaluate the definite integral: $\int_0^1 (5x+1)^3\,dx$.

Final Answer: $\frac{1295}{20}$. Substitute $u = 5x+1 \Rightarrow du = 5\,dx$. Integral: $\frac{1}{5}\left[\frac{u^4}{4}\right]_1^{6} = \frac{(6)^4 - 1^4}{20} = \frac{1295}{20}$.

12 [Intermediate] • Target: Indefinite Integration Techniques

Evaluate the definite integral: $\int_0^1 (6x+1)^3\,dx$.

Final Answer: $\frac{2400}{24}$. Substitute $u = 6x+1 \Rightarrow du = 6\,dx$. Integral: $\frac{1}{6}\left[\frac{u^4}{4}\right]_1^{7} = \frac{(7)^4 - 1^4}{24} = \frac{2400}{24}$.

13 [Intermediate] • Target: Definite Integrals and Area Under Curves

Evaluate the definite integral: $\int_0^1 (7x+1)^3\,dx$.

Final Answer: $\frac{4095}{28}$. Substitute $u = 7x+1 \Rightarrow du = 7\,dx$. Integral: $\frac{1}{7}\left[\frac{u^4}{4}\right]_1^{8} = \frac{(8)^4 - 1^4}{28} = \frac{4095}{28}$.

14 [Intermediate] • Target: Area Between Two Intersecting Curves

Evaluate the definite integral: $\int_0^1 (8x+1)^3\,dx$.

Final Answer: $\frac{6560}{32}$. Substitute $u = 8x+1 \Rightarrow du = 8\,dx$. Integral: $\frac{1}{8}\left[\frac{u^4}{4}\right]_1^{9} = \frac{(9)^4 - 1^4}{32} = \frac{6560}{32}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Rectilinear Motion and Kinematics • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

Evaluate the definite integral: $\int_0^1 (3x+1)^3\,dx$.

Final Answer: $\frac{255}{12}$. Substitute $u = 3x+1 \Rightarrow du = 3\,dx$. Integral: $\frac{1}{3}\left[\frac{u^4}{4}\right]_1^{4} = \frac{(4)^4 - 1^4}{12} = \frac{255}{12}$.

© ClipSAT — clipsat.org | All Rights Reserved

16 [Challenging] • Target: Integration & Kinematics Practice Exercises • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

Evaluate the definite integral: $\int_0^1 (4x+1)^3\,dx$.

Final Answer: $\frac{624}{16}$. Substitute $u = 4x+1 \Rightarrow du = 4\,dx$. Integral: $\frac{1}{4}\left[\frac{u^4}{4}\right]_1^5 = \frac{(5)^4 - 1^4}{16} = \frac{624}{16}$.

17 [Challenging] • Target: Indefinite Integration Techniques • Cross-Lesson: 3.01 Indices, Surds and Polynomials

Evaluate the definite integral: $\int_0^1 (2x+1)^3\,dx$.

Final Answer: $\frac{80}{8}$. Substitute $u = 2x+1 \Rightarrow du = 2\,dx$. Integral: $\frac{1}{2}\left[\frac{u^4}{4}\right]_1^3 = \frac{(3)^4 - 1^4}{8} = \frac{80}{8}$.

18 [Challenging] • Target: Definite Integrals and Area Under Curves • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

Find the area enclosed between the curve $y = 2x^2$, the x-axis, and the lines $x = 0$ and $x = 4$.

Final Answer: 128/3. Area $= \int_0^4 (2x^2)dx = [2/3x^3]_0^4 = 128/3$

19 [Challenging] • Target: Area Between Two Intersecting Curves • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

Find the area enclosed between the curve $y = x^2$, the x-axis, and the lines $x = 0$ and $x = 4$.

Final Answer: 64/3. Area $= \int_0^4 (x^2)dx = [1/3x^3]_0^4 = 64/3$

20 [Challenging] • Target: Rectilinear Motion and Kinematics • Cross-Lesson: 6.01 Permutations and Combinations

Find the area enclosed between the curve $y = 2x^2$, the x-axis, and the lines $x = 0$ and $x = 3$.

Final Answer: 18. Area $= \int_0^3 (2x^2)dx = [2/3x^3]_0^3 = 18$

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Integration & Kinematics Practice Exercises

Evaluate the definite integral: \(\int_0^1 (2x + 1)^3 \, dx \).

a. \(\frac{80}{2} \)

b. \(\frac{84}{8} \)

c. \(\frac{80}{4} \)

d. **\(\frac{80}{8} \)** ✓

Substitute \(u = 2x + 1 \implies du = 2\,dx \). Integral: \(\frac{1}{2}\left[\frac{u^4}{4}\right]_1^{3} = \frac{(3)^4 - 1^{4}}{8} = \frac{80}{8} \).

© ClipSAT — clipsat.org | All Rights Reserved

22 [Foundational] • Target: Integration & Kinematics Practice Exercises

Evaluate the definite integral: \(\int_0^1 (3x + 1)^3 \, dx \).

a. \(\frac{255}{4} \)

b. \(\frac{255}{3} \)

c. \(\frac{259}{12} \)

d. **\(\frac{255}{12} \)** ✓

Substitute \(u = 3x + 1 \implies du = 3\,dx \). Integral: \(\frac{1}{3}\left[\frac{u^4}{4}\right]_1^{4} = \frac{(4)^4 - 1^{4}}{12} = \frac{255}{12} \).

23 [Foundational] • Target: Integration & Kinematics Practice Exercises

Evaluate the definite integral: \(\int_0^1 (4x + 1)^3 \, dx \).

a. \(\frac{625}{16} \)

b. \(\frac{624}{4} \)

c. \(\frac{628}{16} \)

d. **\(\frac{624}{16} \)** ✓

Substitute \(u = 4x + 1 \implies du = 4\,dx \). Integral: \(\frac{1}{4}\left[\frac{u^4}{4}\right]_1^{5} = \frac{(5)^4 - 1^{4}}{16} = \frac{624}{16} \).

24 [Foundational] • Target: Integration & Kinematics Practice Exercises

Evaluate the definite integral: \(\int_0^1 (5x + 1)^3 \, dx \).

a. \(\frac{1295}{5} \)

b. \(\frac{1295}{4} \)

c. **\(\frac{1295}{20} \)** ✓

d. \(\frac{1299}{20} \)

Substitute \(u = 5x + 1 \implies du = 5\,dx \). Integral: \(\frac{1}{5}\left[\frac{u^4}{4}\right]_1^{6} = \frac{(6)^4 - 1^{4}}{20} = \frac{1295}{20} \).

25 [Foundational] • Target: Integration & Kinematics Practice Exercises

Evaluate the definite integral: \(\int_0^1 (6x + 1)^3 \, dx \).

a. \(\frac{2400}{6} \)

b. **\(\frac{2400}{24} \)** ✓

c. \(\frac{2404}{24} \)

d. \(\frac{2400}{4} \)

Substitute \(u = 6x + 1 \implies du = 6\,dx \). Integral: \(\frac{1}{6}\left[\frac{u^4}{4}\right]_1^{7} = \frac{(7)^4 - 1^{4}}{24} = \frac{2400}{24} \).

© ClipSAT — clipsat.org | All Rights Reserved

26 [Foundational] • Target: Integration & Kinematics Practice Exercises

Evaluate the definite integral: \(\int_0^1 (7x + 1)^3 \, dx \).

a. \(\frac{4095}{4} \)

b. \(\frac{4099}{28} \)

c. **\(\frac{4095}{28} \)** ✓

d. \(\frac{4095}{7} \)

Substitute \(u = 7x + 1 \implies du = 7\,dx \). Integral: \(\frac{1}{7}\left[\frac{u^4}{4}\right]_1^{8} = \frac{(8)^4 - 1^{4}}{28} = \frac{4095}{28} \).

27 [Foundational] • Target: Integration & Kinematics Practice Exercises

Evaluate the definite integral: \(\int_0^1 (8x + 1)^3 \, dx \).

a. \(\frac{6560}{4} \)

b. \(\frac{6560}{8} \)

c. \(\frac{6564}{32} \)

d. **\(\frac{6560}{32} \)** ✓

Substitute \(u = 8x + 1 \implies du = 8\,dx \). Integral: \(\frac{1}{8}\left[\frac{u^4}{4}\right]_1^{9} = \frac{(9)^4 - 1^{4}}{32} = \frac{6560}{32} \).

28 [Foundational] • Target: Integration & Kinematics Practice Exercises

Evaluate the definite integral: \(\int_0^1 (9x + 1)^3 \, dx \).

a. \(\frac{9999}{4} \)

b. \(\frac{9999}{9} \)

c. \(\frac{10003}{36} \)

d. **\(\frac{9999}{36} \)** ✓

Substitute \(u = 9x + 1 \implies du = 9\,dx \). Integral: \(\frac{1}{9}\left[\frac{u^4}{4}\right]_1^{10} = \frac{(10)^4 - 1^{4}}{36} = \frac{9999}{36} \).

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Indefinite Integration Techniques

Find the indefinite integral $\int (3x-1)^4\,dx$.

a. $\frac{(3x-1)^5}{15}+C$ ✓

b. $\frac{(3x-1)^5}{5}+C$

c. $\frac{(3x-1)^3}{3}+C$

d. $12(3x-1)^3+C$

Substitute $u=3x-1, du=3dx$: $\int u^4\frac{du}{3}=\frac{u^5}{15}+C=\frac{(3x-1)^5}{15}+C$.

© ClipSAT — clipsat.org | All Rights Reserved

30 [Intermediate] • Target: Indefinite Integration Techniques

Evaluate $\int (e^{2x} + \sin 3x)\,dx$.

a. $\frac{1}{2}e^{2x} - \frac{1}{3}\cos 3x + C$ ✓

b. $\frac{1}{2}e^{2x} + \frac{1}{3}\cos 3x + C$

c. $2e^{2x} - 3\cos 3x + C$

d. $e^{2x} - \cos 3x + C$

$\int e^{2x}dx = \frac{1}{2}e^{2x}$, $\int \sin 3x dx = -\frac{1}{3}\cos 3x$.

31 [Intermediate] • Target: Indefinite Integration Techniques

A curve passes through $(1, 4)$ and satisfies $\frac{dy}{dx} = 4x - 1$. Find the equation of the curve.

a. $y = 2x^2 - x + 3$ ✓

b. $y = 2x^2 - x + 4$

c. $y = 4x^2 - x + 1$

d. $y = 2x^2 + x + 1$

$y = \int (4x - 1)dx = 2x^2 - x + C$. At $(1, 4)$: $2(1) - 1 + C = 4 \Rightarrow C = 3$.

32 [Intermediate] • Target: Definite Integrals and Area Under Curves

Evaluate the exact value of the definite integral $\int_1^4 \left(2x + \frac{1}{\sqrt{x}}\right) dx$.

a. 17 ✓

b. 15

c. 19

d. 16

Antiderivative: $x^2 + 2\sqrt{x}$. At 4: $16 + 2(2) = 20$. At 1: $1 + 2(1) = 3$. Difference: $20 - 3 = 17$.

33 [Intermediate] • Target: Definite Integrals and Area Under Curves

Find the area enclosed by the curve $y = 6x - x^2$ and the x-axis.

a. 36 ✓

b. 18

c. 72

d. 24

Roots: $x(6 - x) = 0 \Rightarrow 0$ and 6. Area $= \int_0^6 (6x - x^2)dx = [3x^2 - x^3/3]_0^6 = 108 - 72 = 36$.

© ClipSAT — clipsat.org | All Rights Reserved

34 [Intermediate] • Target: Definite Integrals and Area Under Curves

Evaluate $\int_0^{\frac{\pi}{4}} \sec^2 x \, dx$.

a. 1 ✓

b. $\sqrt{2}$

c. $\frac{1}{2}$

d. $\pi/4$

$[\tan x]_0^{\pi/4} = \tan(\pi/4) - \tan(0) = 1 - 0 = 1$.

35 [Intermediate] • Target: Area Between Two Intersecting Curves

Find the area of the finite region enclosed between the parabola $y = x^2$ and the line $y = 2x$.

a. $\frac{4}{3}$ ✓

b. $\frac{2}{3}$

c. 2

d. $\frac{8}{3}$

Intersections: $x^2 = 2x \Rightarrow x = 0, 2$. Area $= \int_0^2 (2x - x^2)dx = [x^2 - x^3/3]_0^2 = 4 - 8/3 = 4/3$.

36 [Intermediate] • Target: Area Between Two Intersecting Curves

Find the area between the curves $y = 4 - x^2$ and $y = x^2 - 4$.

a. $\frac{64}{3}$ ✓

b. $\frac{32}{3}$

c. 16

d. 8

Intersections: $4 - x^2 = x^2 - 4 \Rightarrow 2x^2 = 8 \Rightarrow x = \pm 2$. Area $= \int_{-2}^{2} (8 - 2x^2)dx = [8x - 2x^3/3]_{-2}^{2} = (16 - 16/3) - (-16 + 16/3) = 64/3$.

37 [Intermediate] • Target: Area Between Two Intersecting Curves

Set up and evaluate the integral representing the area between $y = \sqrt{x}$ and $y = x$ for $0 \le x \le 1$.

a. $\frac{1}{6}$ ✓

b. $\frac{1}{3}$

c. $\frac{1}{2}$

d. $\frac{2}{3}$

$\int_0^1 (x^{1/2} - x)dx = [\frac{2}{3}x^{3/2} - \frac{x^2}{2}]_0^1 = \frac{2}{3} - \frac{1}{2} = \frac{1}{6}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

© ClipSAT — clipsat.org | All Rights Reserved

38 [Challenging] • Target: Rectilinear Motion and Kinematics • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

A particle moves in a straight line with velocity $v(t) = 3t^2 - 12t + 9$ m/s. Find the displacement from $t = 0$ to $t = 3$.

a. 0 m ✓

b. 9 m

c. 18 m

d. 6 m

$s(3) - s(0) = \int_0^3 (3t^2 - 12t + 9)dt = [t^3 - 6t^2 + 9t]_0^3 = 27 - 54 + 27 = 0$ m.

39 [Challenging] • Target: Rectilinear Motion and Kinematics • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

Find the acceleration of the particle in question 10 at $t = 4$ seconds.

a. 12 m/s^2 ✓

b. 6 m/s^2

c. 24 m/s^2

d. 0 m/s^2

$a(t) = v'(t) = 6t - 12$. At $t = 4$: $a(4) = 24 - 12 = 12$ m/s^2.

40 [Challenging] • Target: Rectilinear Motion and Kinematics • Cross-Lesson: 6.01 Permutations and Combinations

Calculate the total distance travelled by the particle with $v(t) = 3(t - 1)(t - 3)$ from $t = 0$ to $t = 3$.

a. 8 m ✓

b. 0 m

c. 4 m

d. 12 m

Velocity is positive on $(0, 1)$ ($s(1) = 1 - 6 + 9 = 4$) and negative on $(1, 3)$ ($s(3) - s(1) = 0 - 4 = -4$). Distance $= |4| + |-4| = 8$ m.

41 [Challenging] • Target: Integration & Kinematics Practice Exercises • Cross-Lesson: 7.01 Differentiation and Rates of Change

Find the area enclosed between the curve $y = 3x$, the x-axis, and the lines $x = 0$ and $x = 3$.

a. 37/2

b. **27/2** ✓

c. 27

d. 25/2

Area $= \int_0^3 (3x)dx = [3/2x^2]_0^3 = 27/2$

© ClipSAT — clipsat.org | All Rights Reserved

42 [Challenging] • Target: Indefinite Integration Techniques • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

Find the area enclosed between the curve $y = 3x$, the x-axis, and the lines $x = 0$ and $x = 5$.

a. 85/2

b. **75/2 ✓**

c. 73/2

d. 75

Area $= \int_0^5 (3x)dx = [3/2x^2]_0^5 = 75/2$

43 [Challenging] • Target: Definite Integrals and Area Under Curves • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

Find the area enclosed between the curve $y = 3x^2$, the x-axis, and the lines $x = 0$ and $x = 5$.

a. 124

b. 130

c. 250

d. **125 ✓**

Area $= \int_0^5 (3x^2)dx = [x^3]_0^5 = 125$

44 [Challenging] • Target: Area Between Two Intersecting Curves • Cross-Lesson: 3.01 Indices, Surds and Polynomials

Find the area enclosed between the curve $y = 3x^2$, the x-axis, and the lines $x = 0$ and $x = 3$.

a. 54

b. 32

c. 26

d. **27 ✓**

Area $= \int_0^3 (3x^2)dx = [x^3]_0^3 = 27$

45 [Challenging] • Target: Rectilinear Motion and Kinematics • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

Find the area enclosed between the curve $y = x$, the x-axis, and the lines $x = 0$ and $x = 3$.

a. 19/2

b. 9

c. **9/2 ✓**

d. 7/2

Area $= \int_0^3 (x)dx = [1/2x^2]_0^3 = 9/2$

© ClipSAT — clipsat.org | All Rights Reserved